REGIS UNIVERSITY

COLLEGE OF COMPUTER & INFORMATION SCIENCES

Course Syllabus

Course Number: CS 473
Course Title: Artificial Intelligence

Course Description

CS 473. ARTIFICIAL INTELLIGENCE (3). Studies computer based agents that perceive and act rationally within an environment. Introduction to the technologies and algorithms used to construct agents that represent knowledge, search spaces, perform inference, and learn

Prerequisite Courses

CS324 (or CS 372), CS390, MT360A

Course Overview

This course is designed to provide a broad introduction to Artificial Intelligence.

Course Outcomes

Upon completion of this course, students should be able to (in no particular order):

1. Elaborate on fundamental Philosophical, Definitional, and Ethical questions in Intelligent Systems including the Turning and "Chinese Room" thought experiments, rational reasoning, and the risks of developing Artificial Intelligence agents.
2. Determine the characteristics of a given problem that an intelligent system must solve.
3. Explain the defining characteristics, fundamental issues, and fundamental approaches used in: state-space search, knowledge representation & inference, machine learning, reasoning with uncertainty, planning, natural language processing, and intelligent agents.
4. Formulate an efficient problem space for a problem given in English in terms of states, goals, operators, heuristics, including implementing a search algorithm in this space.
5. Formulate an appropriate knowledge representation and inference approach for a problem in terms of predicate logic including implementing this approach.
6. Design and implement an HMM as one example of a temporal probabilistic system.
7. Formulate and apply a simple machine learning approach for a problem given in English including implementing this approach
8. Describe HHM and Neural network approaches to transition-based parsing.

©2017 Regis University

Course Materials

Required Text:

Russell, S, & Norvig, P. (2010). *Artificial Intelligence: A modern approach*. (3rd ed.). Upper Saddle River, NJ: Prentice Hall.

Technology Tools:

1. A PC-compatible computer system running a version of the Windows operating system, or a Mac and administrator rights to install new software.

   http://www.regis.edu/Academics/Learning-Management-System/System-Requirements.aspx

2. NetBeans with Java IDE

Pre-Assignments:

Online Format: Sign on to WorldClass and become familiar with the course navigation of the Web Curriculum

Classroom and Online: Complete the Topic 1 textbook reading Assignment (see below).

Course Assignments and Activities:

15 Week	8 Week	Topics	Readings	Assignments* (Overall course %)
1	1	**1:** Fundamental Issues **2:** Rational Agents	§1.1 Chapter 2	Topic 2 Assignment (2%)
2		**3**: Search Strategies Part I – Classic, Informed & Uninformed search	Chapter 3	Topic 3 Assignment (6%)
3	2	**4**: Search Strategies Part II – – Adversarial Search, Min-max, Alpha-Beta Pruning	Chapter 4 Chapter 5	Topic 4 Assignments (7%)
4		– Constraint Satisfaction Problems	Chapter 6	Exam 1 (10%)
5	3	**5:** Basic Knowledge Representation & Reasoning – Propositional & Predicate Logic	Chapter 7 Chapter 8	Topic 5 Assignment (7%)
6			Chapter 9	
7	4	**6**: Advanced Knowledge Representation & Reasoning – Planning	Chapter 10	Topic 6 Assignments (7%)
8		– Non-classic logics, Sematic Nets	Chapter 12	Exam 2 (10%)
9	5	**7**: Reasoning Under Uncertainty – Probabilistic Reasoning, Bayesian Networks	Chapter 13 Chapter 14	Topic 7 Assignments (7%)
10			Chapter 15	
11	6	**8:** Machine Learning Part I – Classification, Neural Net – Inductive Learning, Bayesian, Decision Trees	Chapter 18	Topic 8 Assignments (7%)
12			Chapter 19	Exam 3 (10%)
13	7	**9:** Machine Learning Part II – – Supervised, Unsupervised	Chapter 20	Topic 9 Assignments (7%)
14		Reinforcement Learning	Chapter 21	
15	8	**10:** AI Applications: Natural Language & Perception **11:** Philosophical & Ethical Issues	§22.1 – 22.3 Chapter 24 Chapter 26	Exam 4 (12%)

* Participation is required each week in classroom or online forum postings (8%)

Summary of Assignments and Percentage Weight towards course grade

Assignment	Value (percent of overall course grade)
Topic Assignments (8 assignments)	50 %
Exams	42 %
Participation	8 %
Course Total	100 %

Late Assignment Policy for Assignments
Late assignments will be graded and then 2% will be deducted for each day the assignment is late, ***up to 5 days late***. Therefore, any assignment turned in more than ***5 days*** late will be given a grade of **zero**, and no feedback will be given.

Additional Assessment

The faculty reserves the right to individually assess a student's knowledge of any course material using additional assessment tools including an oral examination (face-to-face, skype, phone …).

Exams
There will be a midterm and a final exam. Exam questions will be cumulative, taken from reading course content and assignments. Exams will **not** be accepted late.

Participation
Class participation/effort is important because we can all learn from each other. Your participation points can make a difference in the final grade. Participation means:

1. a. Present in class every session (classroom)
   b. Present in the forum every week (online)
2. a. Effectively responds to questions from the facilitator (classroom)
   b. Regularly checks forum and posts all required items by the deadlines (online)
3. Interacts/replies to other students in classroom/forum discussions.

Online Sections: See Faculty Syllabus for Online Weekly Discussion points distribution

Course Policies and Procedures

Adding this course during the Drop/Add Period

If you add this course during the drop/add period, you are responsible for ***immediately*** notifying the instructor that you joined the course late. None of the course due dates will be extended for you. If a due date has already passed when you add the course, late points will be deducted.

Repeating the course

If you are repeating this course (due to a previous withdraw or low grade), you are responsible for ***immediately*** notifying the instructor. If any of the course assignments have not changed since last time you took the course, you may be required to complete alternate assignments.

Plagiarism

Plagiarism includes submitting code obtained from any other person, publication, or any internet web source. ***All work submitted in CS473 must be your own.***

In cases of suspected cheating or plagiarism, the instructor will discuss the matter with the student(s) involved. The instructor reserves the right to question any student orally or in writing about any assignment, and to use the evaluation of the student's understanding of the assignment and of the submitted solution as evidence of cheating. All cheating incidents will be reported to the Computer Science department, and may also be reported to the Academic Integrity Board for further action.

CC&IS Grading Scale

Letter Grade	Percentage	Grade Point
A	93 to 100	4.00
A–	90 to less than 93	3.67
B+	87 to less than 90	3.33
B	83 to less than 87	3.00
B–	80 to less than 83	2.67
C+	77 to less than 80	2.33
C	73 to less than 77	2.00
C–	70 to less than 73	1.67
D+	67 to less than 70	1.33
D	63 to less than 67	1.00
D-	60 to less than 63	.67
F	Less than 60	0

Additional information about grading can be found in the latest edition of the University Catalog, available at http://www.regis.edu/Academics/Course%20Catalog.aspx.

CC&IS Policies and Procedures

Each of the following CC&IS Policies & Procedures is incorporated here by reference. Students are expected to review this information each term, and agree to the policies and procedures as identified here and specified in the latest edition of the University Catalog, available at http://www.regis.edu/Academics/Course%20Catalog.aspx or at the link provided.

- The CC&IS Academic Integrity Policy.
- The Student Honor Code and Student Standards of Conduct.
- Incomplete Grade Policy, Pass / No Pass Grades, Grade Reports.

- The Information Privacy policy and FERPA. For more information regarding FERPA, visit the U.S. Department of Education.
- The HIPPA policies for protected health information. The complete Regis University HIPAA Privacy & Security policy can be found here: http://www.regis.edu/About-Regis-University/University-Offices-and-Services/Auxiliary-Business/HIPAA.aspx.
- The Human Subjects Institutional Review Board (IRB) procedures. More information about the IRB and its processes can be found here: http://regis.edu/Academics/Academic-Grants/Proposals/Regis-Information/IRB.aspx.

The CC&IS Policies & Procedures Syllabus Addendum summarizes additional important policies including, Diversity, Equal Access, Disability Services, and Attendance & Participation that apply to every course offered by the College of Computer & Information Sciences at Regis University.

A copy of the CC&IS Policies & Procedures Syllabus Addendum can be found here: https://in2.regis.edu/sites/ccis/policies/Repository/CCIS%20Syllabus%20Addendum.docx.